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NPS/002/010 – Technical Specification for Jointing Sundries

1. Purpose

The purpose of this document is to detail the specific requirements of NEDL and YEDL in relation to the products detailed within this document.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
MS-U-23		Soldering Sockets and Ferrules
DSS/005/007		Specification for heat shrinkable moulding, sleeving and end caps.
DSS/005/002		Specification for plumbing Metals and Solder

2. Scope

This document applies to the supply of a range of miscellaneous items of a jointing nature intended for use on the companies' distribution network.

3. Requirements

3.1 Cable End Caps

A range of heat shrink and/or cold shrink end caps used to seal the ends of plastic and paper insulated cables from moisture, dirt and corrosion.

Heat-shrink end caps shall be coated with a hot melt adhesive which shall seal onto the cable sheath and core insulation during insulation. The cold shrink breakouts shall contain a suitable sealing medium, to seal onto the sheath of the cable.

The range of cables for which end caps are required are as detailed in Appendix 1.

3.2 Wraparound Sheath Repair

Utilized to affect a permanent repair on the damaged sheath of a buried cable where it would be impossible to install a tube.

The inner wall of the wraparound sleeve shall be coated with a hot melt adhesive. Upon heating the sleeve shall shrink and the adhesive shall melt, thereby providing an insulating and a permanent moisture seal around the cable sheath.

The range of cables for which the wraparound sleeve is required are as detailed in Appendix 1.

3.3 Thin Wall Heat-shrinkable Tubing

A flexible cross-linked, UV resistant, polyolefin thin walled heat-shrinkable tubing utilized for insulating and sealing low-voltage accessories and power cables. The tubing shall have a Shrink ration 3:1

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3.4 Heavy Wall Heat-shrinkable Tubing

A flexible cross-linked, UV resistant, polyolefin heavy walled heat-shrinkable tubing utilized for insulating and sealing low-voltage accessories and power cables. The tubing shall have a Shrink ration 3:1

3.5 Cable Breakouts

A range of cold shrink or heat-shrink cable breakouts required to seal the crotches of plastic or paper-insulated cables. Heat-shrink breakouts shall be coated with a hot melt adhesive which shall seal onto the cable sheath and core insulation during insulation. Cold shrink breakouts, shall contain a suitable sealing medium, to seal onto the cable sheath and core insulation.

Breakouts are required for two, three and four core cables, the details of which are detailed in Appendix 1.

3.6 Transformer Lv Insulating Shrouds

Single and double insulating shrouds, moulded in a thermo plastic or similar material shall be a one-piece snap-on fit and designed to fit the low voltage terminal bushing of a transformer not fitted with a termination chamber.

The single shroud shall be suitable to accommodate a 4-holed palm mechanical lug with the double shroud suitable for 2 x 4-holed palm mechanical lug. Both shrouds shall have a tapered cable outlet which can be cut to suit the different diameters of conductors to be used.

3.7 Fuse Handle

The fuse handle shall be capable of accepting BS88 "J" type fuses. The handle shall be manufactured in glass reinforced polyester or a similar material and the operation shall be by wedge tightening thumb screws.

3.8 Solders & Fluxes

A range of solders and fluxes as detailed: -

- a) Abrasion solder – Tinning of Aluminium sheaths and solid conductors
- b) Grade 'G' soft solder - for Copper conductors
- c) Grade 'H' solder – for Aluminium sheath plumbing
- d) Grade 'D' solder – for lead Sheath plumbing
- e) Grade 'M' solder – for Copper conductors
- f) 16SWG resin cored (5kg reels)
- g) Flux for stranded or solid Aluminium conductors
- h) Flux for stranded Copper conductors
- i) Tallow – Cooling of lead plumbs/lubrication

For items a – e the grade letters shall be endorsed during the moulding process at each end and the middle of the bar. Solders shall comply with the requirements of BS EN29453 or equivalent

3.9 Miscellaneous Items

- Bicaseal Putty 0.5Kg packs
- 16SWG tinned copper wire x .4Kg reels
- Linen thread wax free x 1.5Kg reels
- Antimonial lead wire 2.8mm x 2mm dia.
- 18SWG tinned copper wire x .4Kg reel
- 20SWG tinned copper wire x .4Kg reels

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4. References

4.1 External Documentation

Reference	Title
ENATS 09-11 Issue 1	Heat shrink insulating materials for use with 600/1000V cable and accessories.
BS EN 29453:1994	Specification for soft solder alloys. Chemical compositions and forms.

4.2 Internal documentation

Reference	Title
NIL	

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	07/04/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Procurement Manager	G Bolden	23/05/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D M Gazda	Underground Standards Manager	D M Gazda	23/05/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	02/06/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	23/05/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	25/05/06

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Appendix 1 – Cable Diameters / Requirements etc.

Spec

Number:

3.1 Cables with an approximate outer overall diameter range of between 12mm² - 65mm²

3.2 Cables with an approximate outer overall diameter range of between 28mm² - 65mm²

3.3 Requirements for Thin Walled Tubing: -

- 25mm expanded internal diameter colour **BLACK** x 75M spools
- 25mm expanded internal diameter colour **BLUE** x 75M spools
- 19mm expanded internal diameter colour **BLACK** x 75M spools
- 12mm expanded internal diameter colour **BLACK** x 75M spools

3.4 Requirements for Heavy Walled Tubing: -

- 51mm expanded internal diameter colour **BLACK**
- 43mm expanded internal diameter colour **BLACK**
- 33mm expanded internal diameter colour **BLACK**
- 20mm expanded internal diameter colour **BLACK**

Spool lengths to be agreed

3.5 Cable Breakouts: -

- 2-core for cable range 25mm² – 35mm²
- 3-core for cable range 4mm² - 25mm²
- 3-core for cable range 95mm² – 300mm²
- 4-core for cable range 70mm² – 150mm²
- 4-core for cable range 185mm² – 300mm²

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Appendix 2 – Addendum to Supplier Requirements

To enable NEDL/YEDL to install the product(s) in accordance with the manufacturer's recommendations, if appropriate, the supplier shall provide a single copy of drawings, descriptive leaflets and instruction manuals appropriate to the goods being offered, which shall incorporate details of the recommended installation procedures required in order to provide optimal performance of the goods during their operational life.

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Appendix 3 – Logistical Requirements

The supplier shall ensure that each item is suitably packaged and protected to maintain the product and packaging as “fit for service” prior to installation taking account of the potential for an outdoor storage environment.

All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Suppliers shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the supplier is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included: -

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * NEDL/YEDL brand or name
- * NEDL/YEDL product code
- * Weight

To enable NEDL/YEDL to store the product(s) in accordance with the manufacturer's recommendations the supplier should provide details of the recommended storage environment with respect to each tendered product.